

Structural Design and Performance Evaluation of the GRATTIS Payload Mounting to Support Sensor Performance

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Introduction

GRATTIS

The Gravitational Reference Advanced Technology Test In Space (GRATTIS) mission will **demonstrate the end-to-end functionality and sensitivity performance** of the Simplified Gravitational Reference Sensor (S-GRS) [1].

- **Flies two identical S-GRS** to recover the acceleration sensitivity (noise floor)
- Mounted **symmetrically to the center of mass** (CoM) of an ESPA-class commercial microsatellite
- Planned launch in **early 2027**

S-GRS

The S-GRS is an **ultra-precise electrostatic inertial sensor** for future Earth geodesy missions following the successful CHAMP [2], GRACE [3], GOCE [4], and GRACE-FO [5] missions.

- Designed to **measure non-gravitational forces** and enhance the precision of gravity field recovery
- Estimated to be **10 times more sensitive** than the GRACE accelerometers and more than **500 times more sensitive** if operated on a drag compensated platform [6]

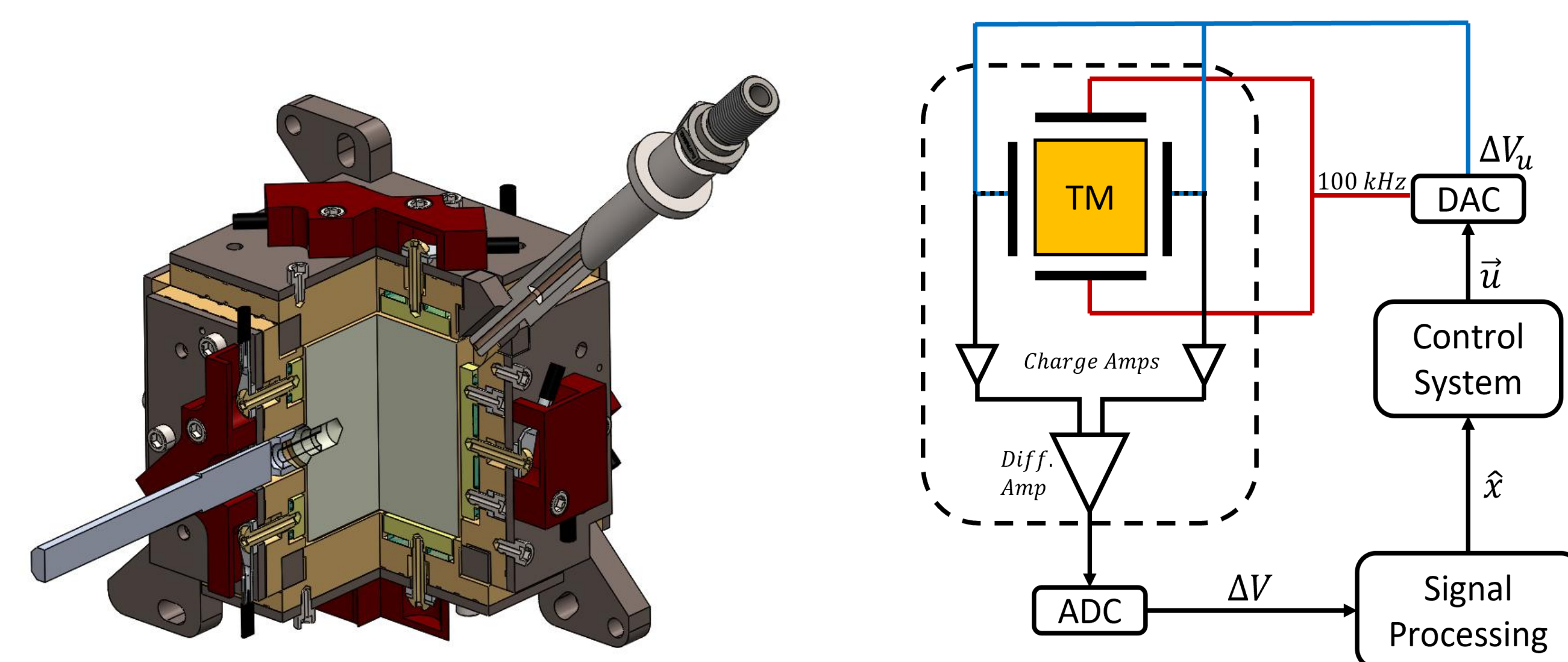


Figure 1: GSE-modified S-GRS head internal view, showing mounting, electrodes, and UV LED charge management.

Figure 2: Sensing and actuation scheme of the S-GRS. Black traces represent capacitive data streams, blue represent actuation potential, and red represent injection signals.

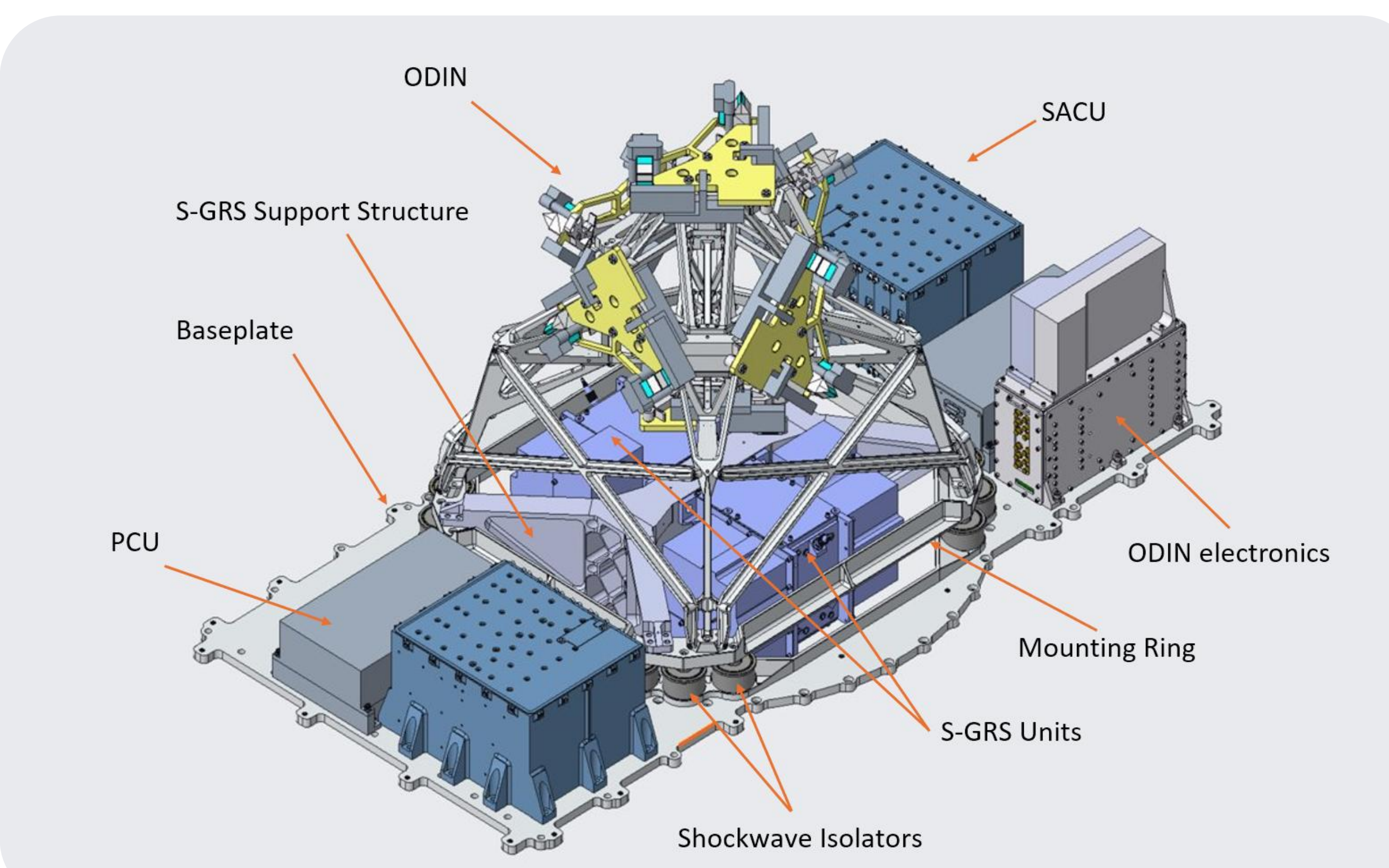


Figure 3: Current CAD assembly of the GRATTIS payload including two S-GRS sensors, the ODIN optomechanical accelerometer package from the University of Arizona, and necessary electronics equipment for the UV LED charge management system, S-GRS, and ODIN. The baseplate will be integrated on an Apex Space Ares class spacecraft bus.

Payload Development

Design

Apex Space provides the bus, while **PSSL integrates instrumentation**. GRATTIS needs to fly two S-GRS units.

- The S-GRS units must be close to the spacecraft CoM and each other **to reduce residual rotational accelerations** (Figure 4)
- A **CTE mismatch from the spacecraft mounting** plate to S-GRS frame must be controlled (Figure 4)
- Structure must survive launch loads and Apex requirements

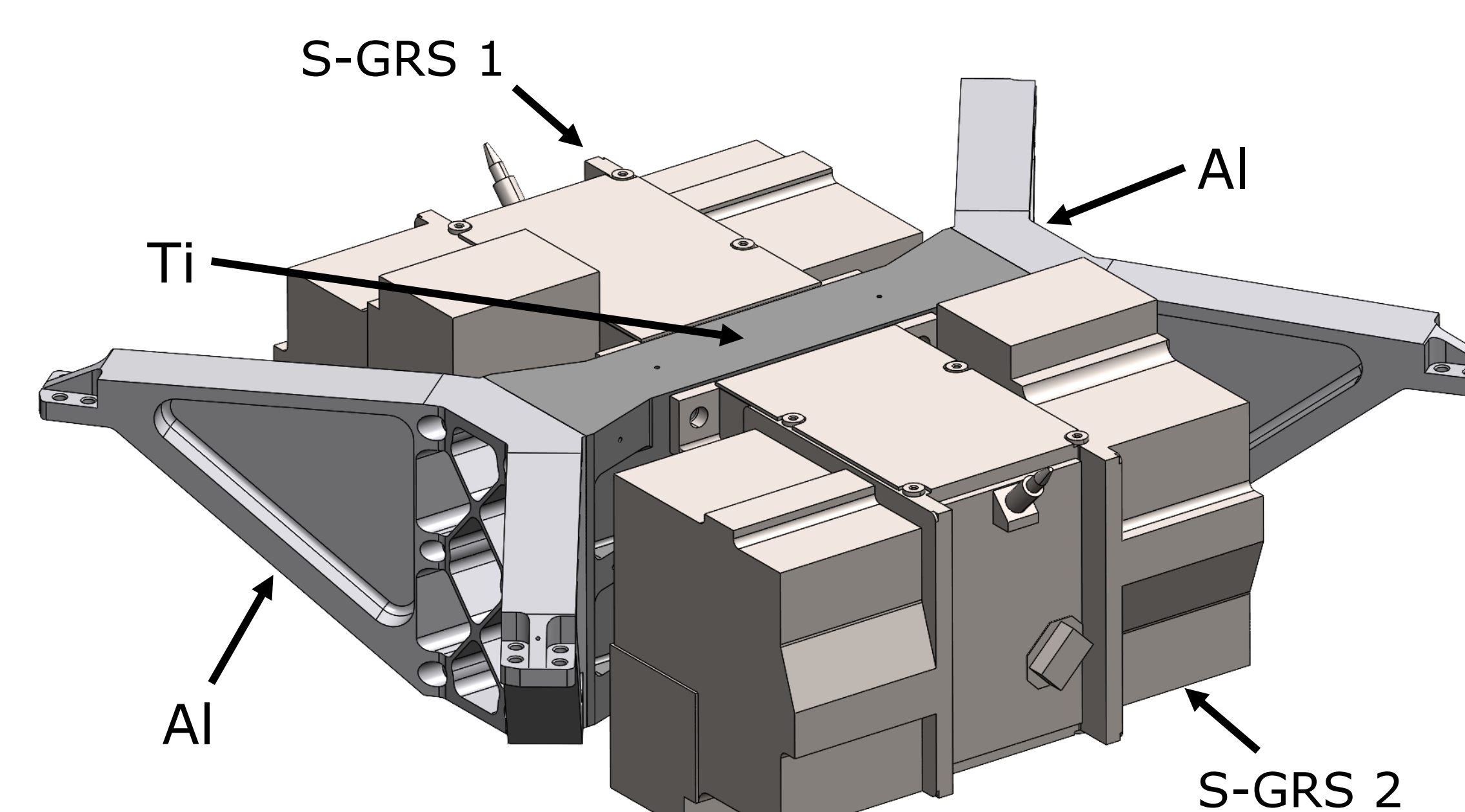


Figure 4: S-GRS support structure to suspend the sensor packages close to the spacecraft CoM and minimize vibrational and thermal disturbances.

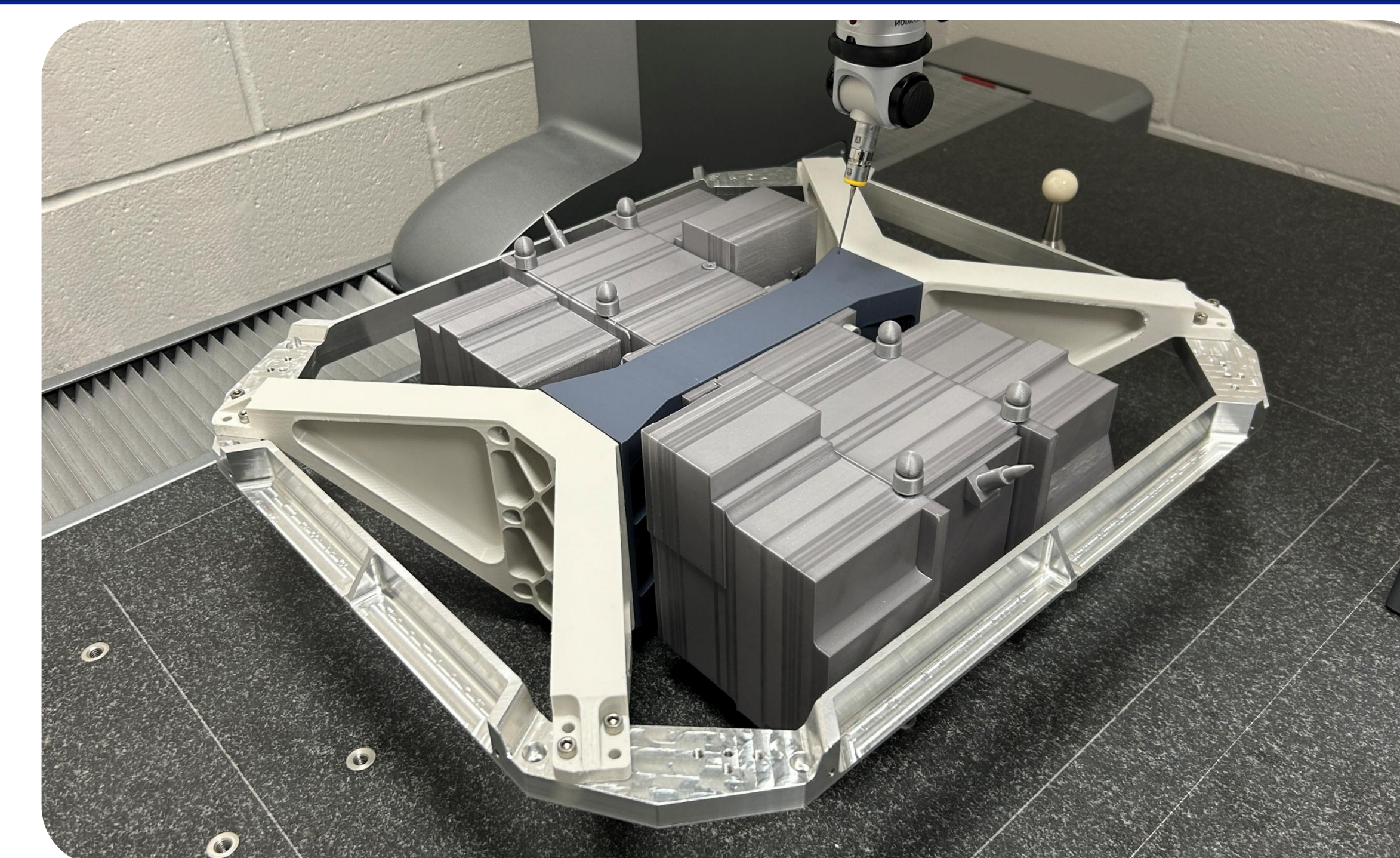


Figure 5: 3D prints of the S-GRS units and support structure mounted to an Al ring on a Hexagon CMM to practice payload metrology.

Metrology and Validation

Pre-flight metrology is performed at UF with a Hexagon CMM.

- External SMRs on the S-GRS units provide **relative position datums for sensor test mass centers**
- **Representative models** are being prepared to develop metrology procedures for final payload metrology (Figure 5)
- Payload metrology will provide an **initial estimate for test mass center-to-center measurements**

Principle of Measurement

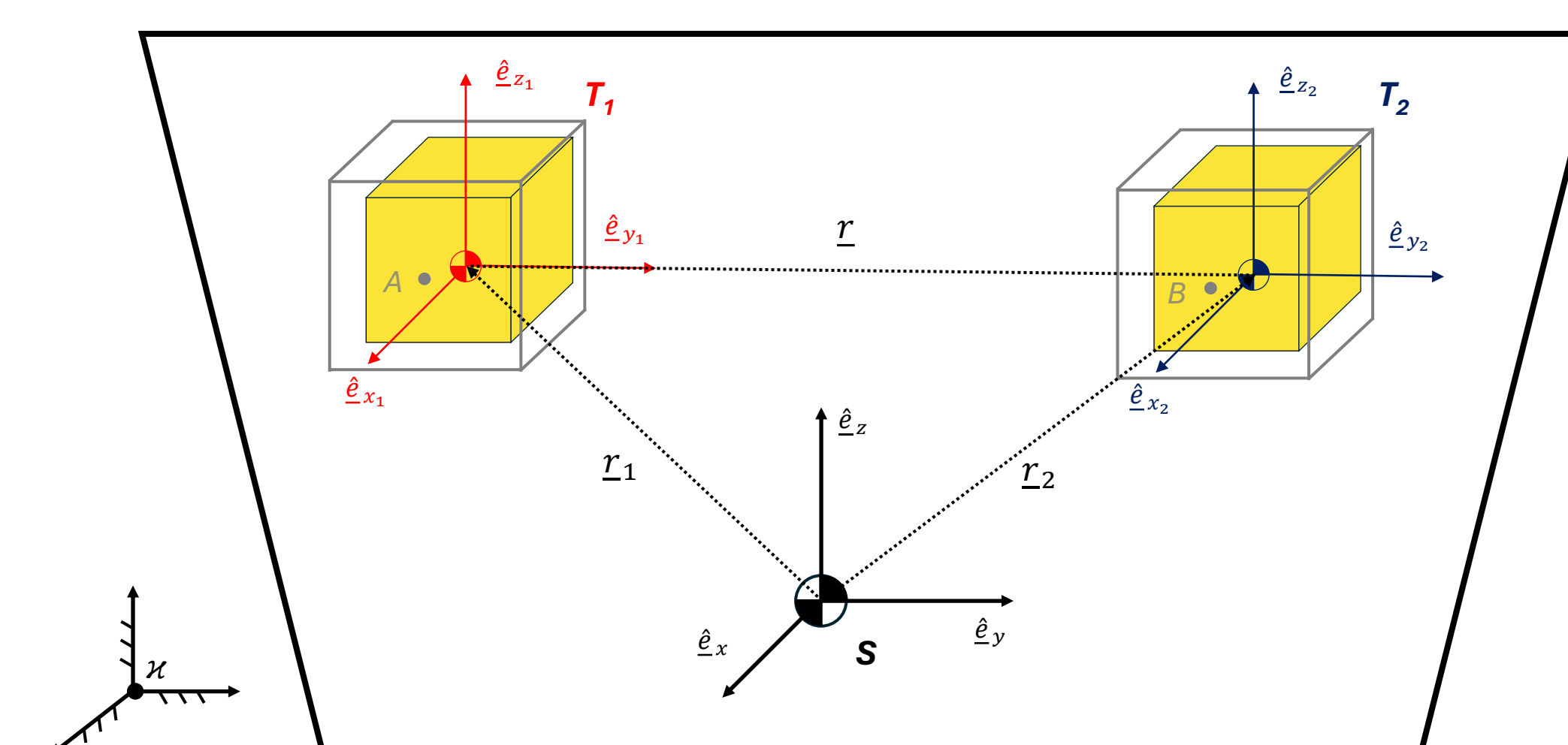


Figure 6: Kinematics diagram for two-accelerometer GRATTIS payload arrangement to model rotational accelerations.

Noise Sensitivity to Test Mass Separation

- Rotation between frame S and T_1 or T_2 is ω_{SC} , so each **acceleration of a TM** in the S frame (Figure 6) is:

$$\underline{a}_i = \ddot{\underline{r}}_i + \underline{\omega}_{SC} \times \underline{r}_i + \underline{\omega}_{SC} \times \dot{\underline{r}}_i + \underline{\omega}_{SC} \times (\dot{\underline{r}}_i + \underline{\omega}_{SC} \times \underline{r}_i) + \underline{n}_i$$
- Differencing the two and grouping terms:

$$\underline{a}_1 - \underline{a}_2 = (\ddot{\underline{r}}_1 - \ddot{\underline{r}}_2) + \underline{\omega}_{SC} \times (\underline{r}_1 - \underline{r}_2) + 2(\underline{\omega}_{SC} \times (\dot{\underline{r}}_1 - \dot{\underline{r}}_2)) + \underline{\omega}_{SC} \times (\underline{\omega}_{SC} \times (\underline{r}_1 - \underline{r}_2)) + \sqrt{2}(\underline{n}_1 - \underline{n}_2)$$
- Measured acceleration on the spacecraft should be the same. With $\underline{r} = \underline{r}_1 - \underline{r}_2$, one gets the **noise difference**:

$$\sqrt{2}\underline{n} = \ddot{\underline{r}} + \underline{\omega}_{SC} \times \underline{r} + 2(\underline{\omega}_{SC} \times \dot{\underline{r}}) + \underline{\omega}_{SC} \times (\underline{\omega}_{SC} \times \underline{r})$$
- $\underline{r}$ **must be minimized** and known to high accuracy

Metrological Precision

Euler Contributions to the Noise Floor

- **Noise goal** is $\leq 10^{-11} \text{ m/s}^2/\sqrt{\text{Hz}}$ at 1 mHz
- Relative positioning must be **known to 10 μm or better**
- Coriolis and centripetal terms are negligible
- Acceleration noise is $|\delta\dot{\omega}_{SC}| = 10^{-6} \text{ rad/s}^2/\sqrt{\text{Hz}}$ and $|\delta\dot{\underline{r}}| \leq 10 \mu\text{m}$ then $\dot{\omega}_{SC} \times \underline{r} \leq 10^{-11} \text{ m/s}^2/\sqrt{\text{Hz}}$
- Relative positioning must be **understood and ensured through design**, then **estimated after launch** and during operation to meet mission objectives (Figure 7)

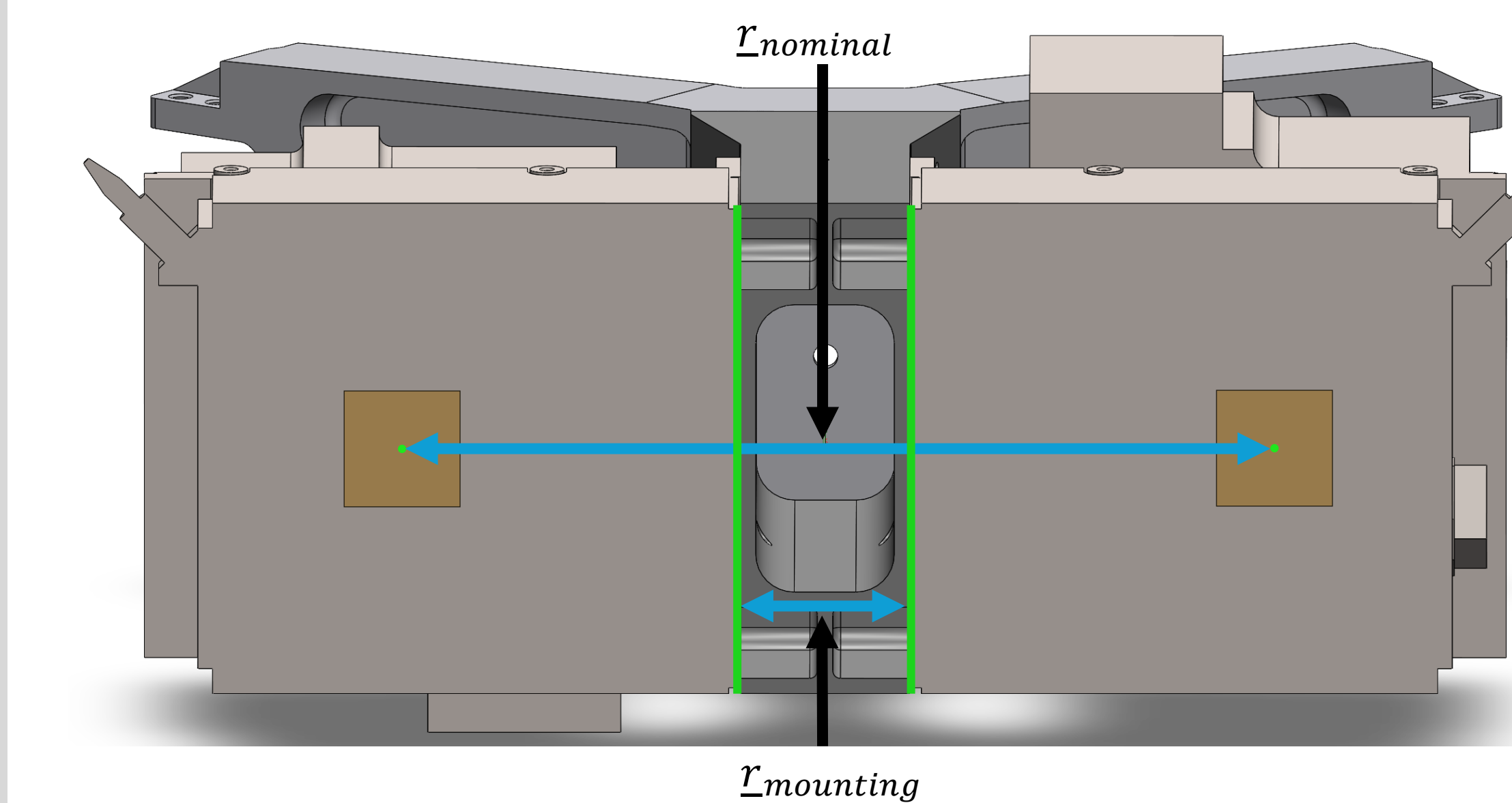


Figure 7: Nominal S-GRS test mass separation and minimum mounting distance constrained by modal and structural requirements for launch.

Metrology Results

Test Mass Metrology

	X	Y	Z
D	6 5 7 0 Δ 0 -3 -3 7	1 13 17 2 5 4 3 9 4	22 18 -15 -5 3 -9 2 7 3
E	6 0 0 4 Δ -4 1 0 -5	-1 -1 2 0 0 4 0 1 3	-8 -7 -4 -7 -7 -5 -2 -6 -4
F	4 8 8 -2 Δ 2 -7 -2 2	-1 0 0 -3 1 -5 -6 -6 -7	-5 -2 3 -4 0 4 -2 1 6

Measurement	TM D	TM E	TM F
X [mm]	30.007	30.009	30.014
Y [mm]	29.998	30.006	30.003
Z [mm]	29.995	29.995	30.001
Mean Deviation [um]	5	7	6
Top Flat [A] [um]	3	8	2
Bottom Flat [um]	4	4	5
Right Flat [B] [um]	3	3	2
Left Flat [um]	5	3	3
Front Flat [C] [um]	5	4	2
Back Flat [um]	4	7	5
Mean Deviation [um]	4	5	3
Right Perp to [A] [um]	16	10	16
Left Perp to [A][C] [um]	20	14	14
Front Perp to [A][B] [um]	18	8	18
Back Perp to [A][B] [um]	15	11	14
Mean Deviation [um]	17	11	16
X Parallel [um]	8	16	15
Y Parallel [um]	20	9	10
Z Parallel [um]	17	6	13
Average Parallel [um]	15	10	13
Mean Deviation [um]	10	8	9

Figure 8: Metrology results using point-to-point measurements across opposing faces for 3 test masses (left). CMM metrology results for the same TMs to confirm measurements and assess GD&T features (right).

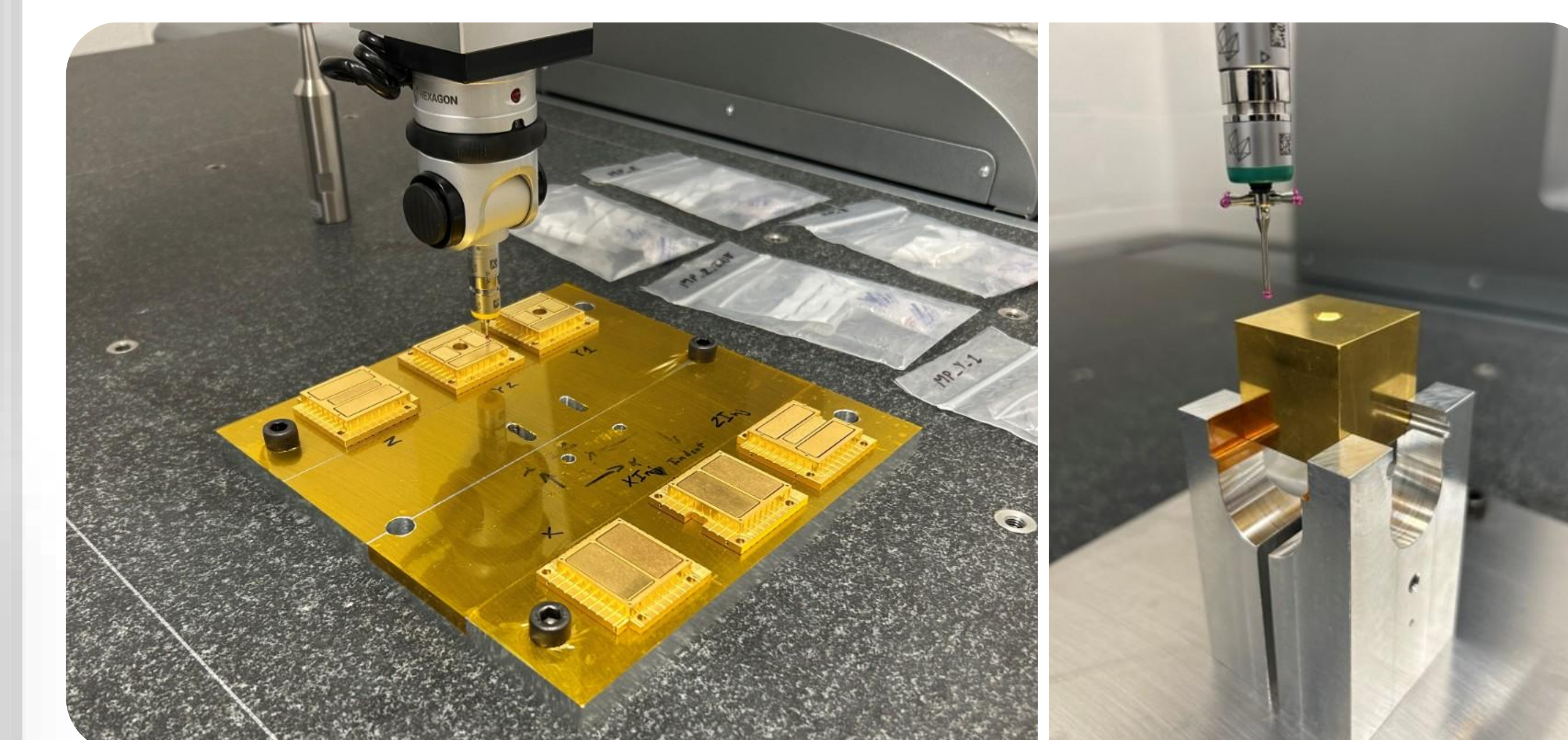


Figure 9: Metrology for the S-GRS electrodes (left) and metrology fixture with a test mass used for Figure 8b results (right).

Summary

- The GRATTIS mission will demonstrate ultra-precise inertial sensing with two high-sensitivity S-GRS units on a microsatellite.
- Differential measurements between accelerometers cancels common noise, recovering acceleration sensitivity.
- Achieving the noise floor goal requires knowing relative sensor positioning within 10 micrometers, motivated by rotational contributions.
- Payload support structure enables precise control of sensor separation and modelling will predict the thermal and structural environment.
- Metrology with a hexagon CMM confirmed initial test mass measurements and show 3 test masses with <10 micrometer mean deviation from ideal shape.

References

- [1] Conklin, J W et al. 2025 [4] Drinkwater et al. 2003
 [2] Reigber C, Lühr H, Schwintzer P 2002 [5] Sheard et al. 2012
 [3] Tapley et al. 2004 [6] Dávila Álvarez, A et al. 2022